PT 100 THERMOMETER PROJECT

IIT HYDERABAD

EE1010

MATHS FOR ELECTRICAL ENGINEERING

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Block Diagram:

PT 100 SENSOR
↓
MICRO CONTROLLER
(ARDUINO UNO)
↓
TEMPERATURE CALCULATION
(RESISTANCE TO TEMP CONVERSION)
↓
TEMP DISPLAY LCD

Objective:

To design and develop a PT100-based digital thermometer using an Arduino Uno and a 16 × 2 LCD display to measure temperature by detecting changes in the resistance of the PT100 sensor, processing the corresponding electrical signal, and displaying the measured temperature in real time.

1.Components and their functions:

Component	Function
Arduino uno	Reads the sensor signal , calculates temperature, and controls the lcd.
Bread board	Helps to connect and test the circuit without soldering.
Perf board	Holds the components for permanent circuit connections using soldering.

Jumper wires	**Connect the components electrically.**
Pt 100	**Senses temperature by changing its electrical resistance.**
Lcd display	**Displays the measured temperature .**
Potentiometer	**Adjust the contrast of the lcd display.**

2.Progress log of thermometer:

2.1 Lcd test

a) **Connect 5V and GND of Arduino to power rails of bread board.**
b) **Lcd connections:**

lcd pin 1 - GND

lcd pin 2 – 5V

lcd pin 3 – Middle pin of potentiometer

lcd pin 4 – D12

lcd pin 5 – GND

lcd pin 6 – D11

lcd pin 7 to 10 - no connections

lcd pin 11 – D5

lcd pin 12 – D4

lcd pin 13 – D3

lcd pin 14 – D2

lcd pin 15 – 5V

lcd pin 16 – GND

b) The potentiometer was connected between the Arduino's 5V and GND.

c) Initially, the LCD was tested by uploading a test program to Arduino, and all connections were verified to ensure proper functionality.

d) The LCD worked properly when the Arduino was powered through a USB cable, displaying the message "PT100 Thermometer – LCD Test Okay", confirming that the LCD and its connections were functioning correctly.



2.2 Testing with bread board

a) the connections between Arduino, lcd and potentiometer are taken as above (lcd testing connections)

b) One of the resistor pin is connected to GND and the other is connected to A0 of the Arduino.

c)Two white pins (with less resistance difference) of pt100 are connected to 5V and red pin of pt100 is connected to the resistor pin which is connected A0 of Arduino.

d)At first we used linear equation(calibration Equation) In the code.

e) Calibration Equation Improvement**: Initially, a linear equation was used in the code to estimate the temperature from the PT100 sensor readings. However, to improve measurement accuracy and account for the nonlinear relationship between the sensor resistance and temperature, the linear equation was later replaced with a quadratic equation. This modification helped provide more accurate temperature readings over the operating range of the thermometer.**

f) We tested this pt100 working using breadboard with hot water, cool water and normal water.

g) We got an error of approximately 10 degrees with that code using quadratic equation.

h) This indicates a need for further calibration and improvement of the temperature conversion formula to achieve more accurate and reliable measurements.

i) After taking new readings with the updated coefficients, the code was adjusted again based on that data.

j) We got error of around 6 degrees with the updated code.

k) After confirming that all components are working properly using lcd test and pt100 temperature cheking test using breadboard, we started soldering them to the perf board.

2.3. Soldering and the final assembly of pt100 digital thermometer.

a) At first, two strips of header pins are soldered to the perf board.

b) One strip is for Lcd and other is for Arduino connections.

c) We soldered two header pins to create a 5V junction, so that all 5V connections can be soldered there.

c) The pins of the strips are soldered as follows,

c.1) Connections of the strip which is to be connected to Lcd.

Lcd pin 1 – GND (pin 1 of other strip)

We connected pin 1 of other strip using a small wire to some random point in the perf board to create a GND junction.

Lcd pin 2 – 5V (pin 2 other strip)

Lcd pin 3 – Middle pin of potentiometer

Lcd pin 4 – D2(pin 4 of other strip)

Lcd pin 5 – GND (pin 1 of other strip)

Lcd pin 6 – D3(pin6 of other strip)

Lcd pins 7 – 10 – no connections

Lcd pin 11 – D4(pin11 of other strip)

Lcd pin 12 – D5(pin 13 of other strip)

Lcd pin 13 – D6(pin 14 of other strip)

Lcd pin 14 – D7(pin 15 of other strip)

Lcd pin 15 – GND(connected to the above mentioned 5V junction)

Lcd pin 16 – 5V(connected to ground 0V junction)

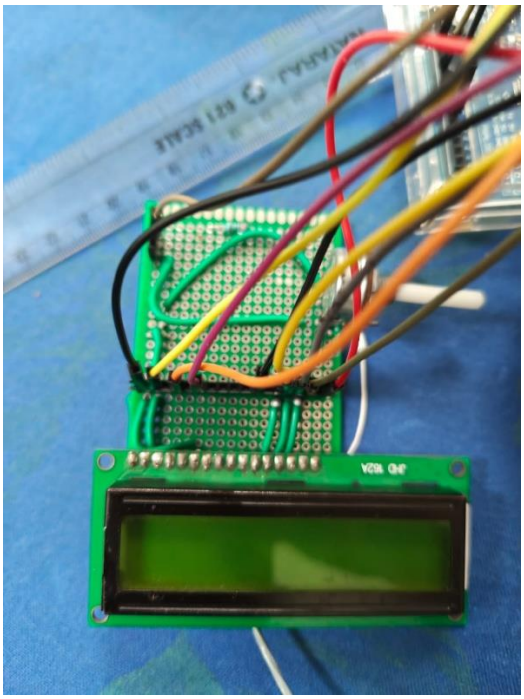
c.2) Last pin of the other strip is connected to A0 of Arduino.

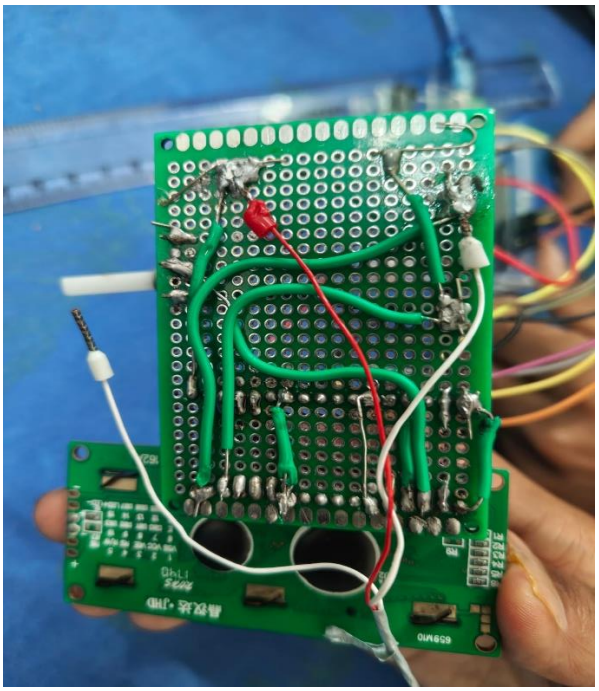
d) The pins of potentiometer are soldered to GND and 5V.

e) One pin of the resistor is connected to GND junction, other Is connected to last pin of second strip which is connected to A0 of Arduino and red pin of pt100.

f) White pin of pt100 is connected to 5V junction.

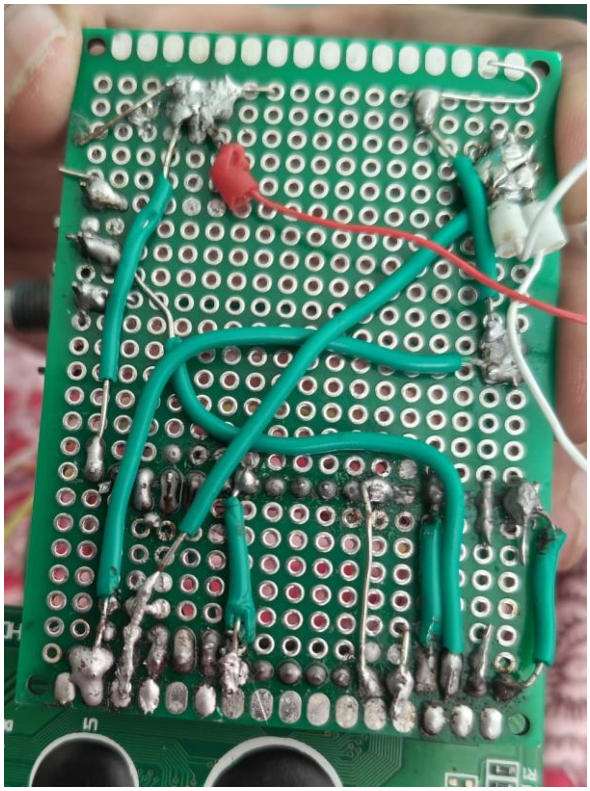
g) After completing soldering part, we connected Arduino's 5V, GND, digital and analog pins to the second strip and 5V junction using jumper wires.





h) ERROR: when we uploaded the code into Arduino, we noticed that backlight is not glowing in Lcd display even

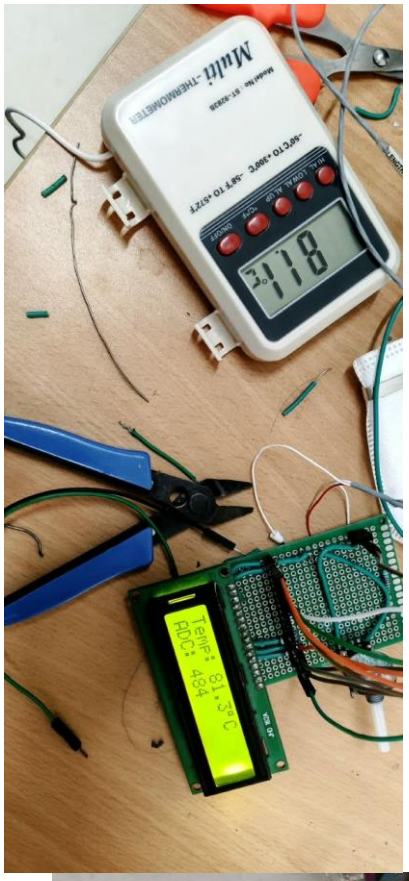
though Lcd display is showing temperature (After rotating potentiometer knob).

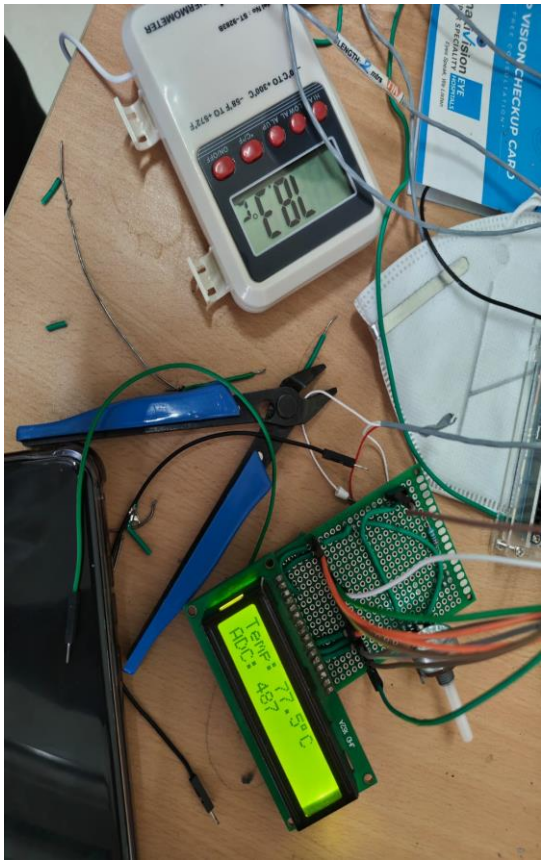


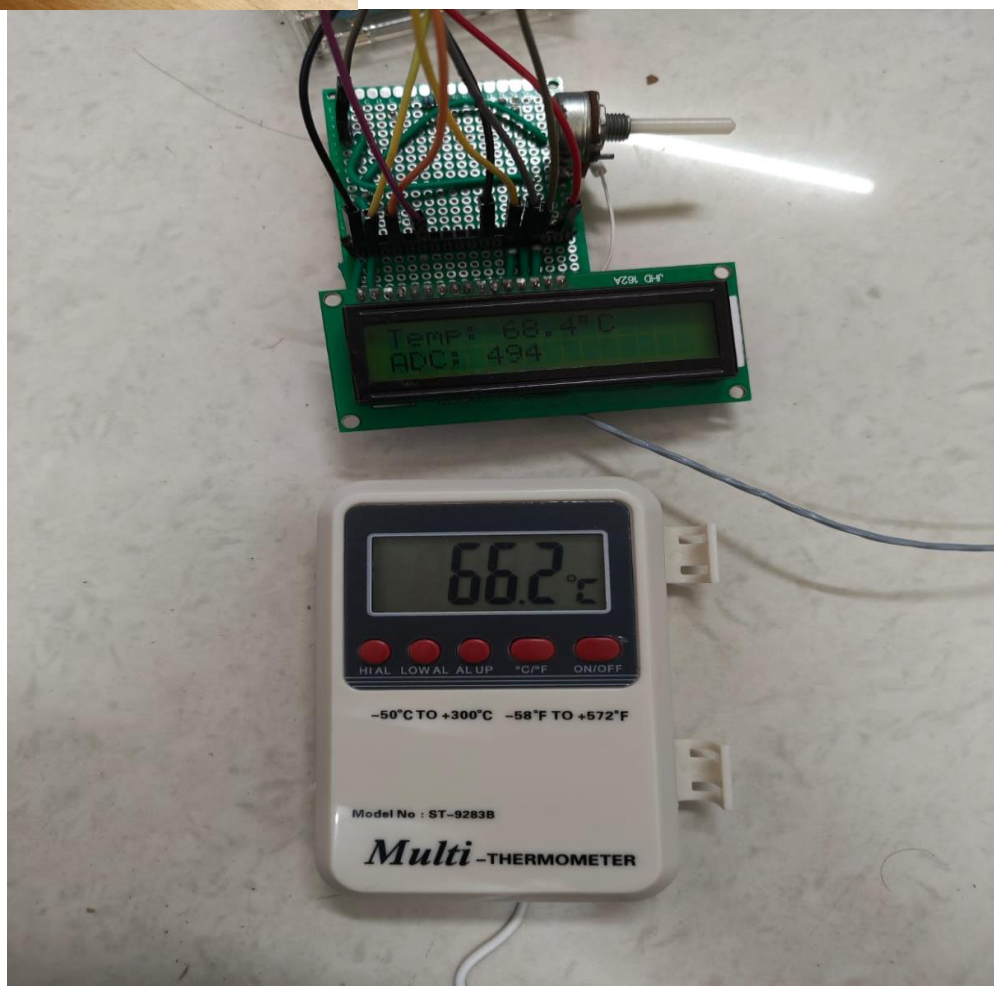
i) The reason for this is incorrect connections of Pins 15, 16.

j) After realizing that the pins 15 and 16 of strip one (which is connected to Lcd) are connected in the reverse manner, we desoldered and made correct connections.

k) We tested the thermometer using water with different temperatures.


Multi -THERMOMETER
Temp: 81.3°C
ADC: 484


Temp: 77.5°C
ADC: 487

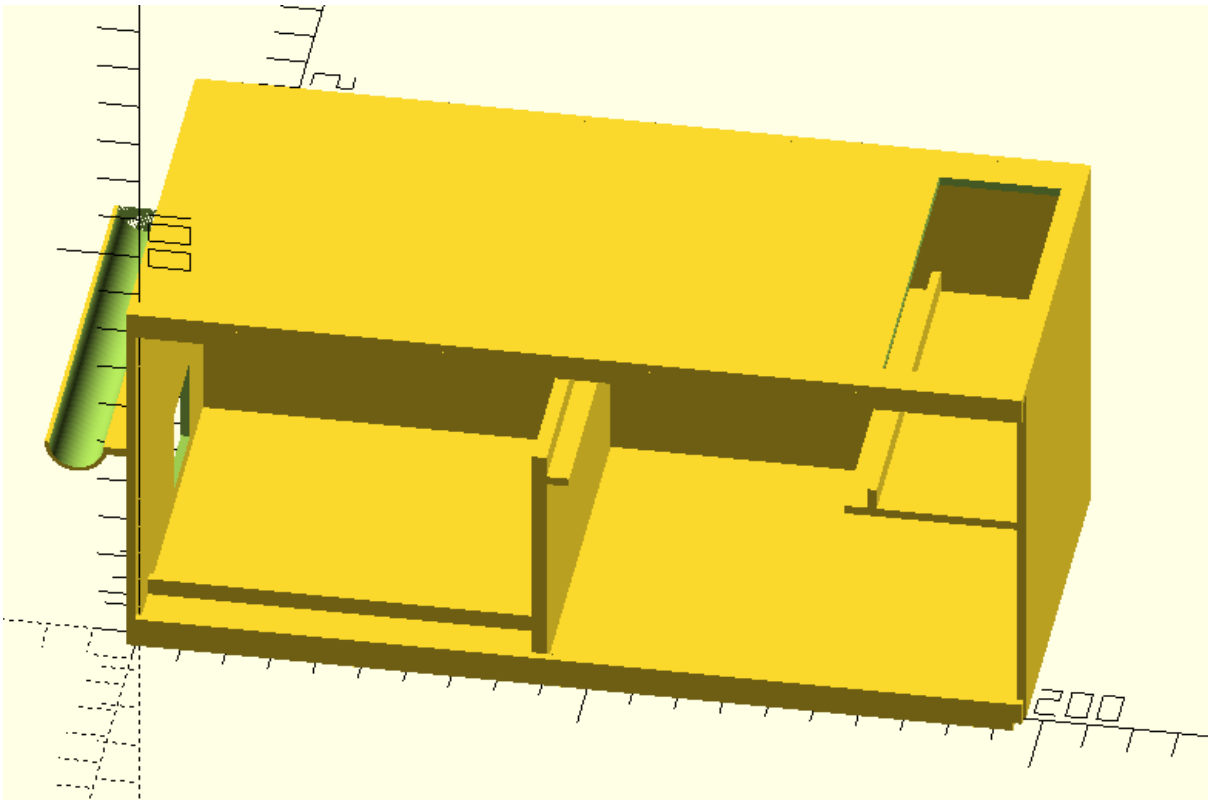

Temp: 68.4°C
ADC: 494
66.2°C
HI AL LOW AL AL UP °C/°F ON/OFF
-50°C TO +300°C -58°F TO +572°F
Model No : ST-9283B
Multi -THERMOMETER

3. Progress log of 3d printing:

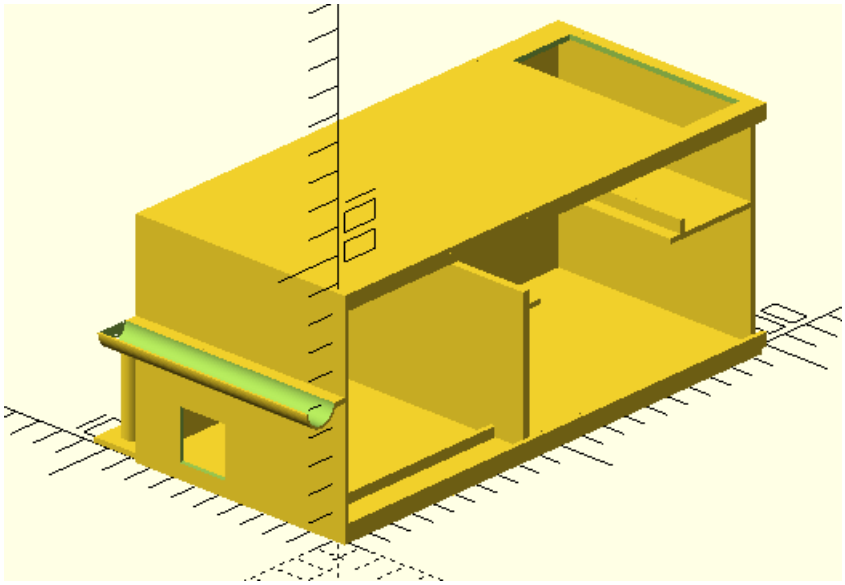
a) The cuboid enclosure is divided into two compartments: one for the Arduino Uno and the other for the perfboard and its mounted components."

b) The left compartment houses the Arduino Uno, while the right compartment is for the perfboard. A cutout is provided at the bottom-left of the enclosure wall for USB cable insertion.

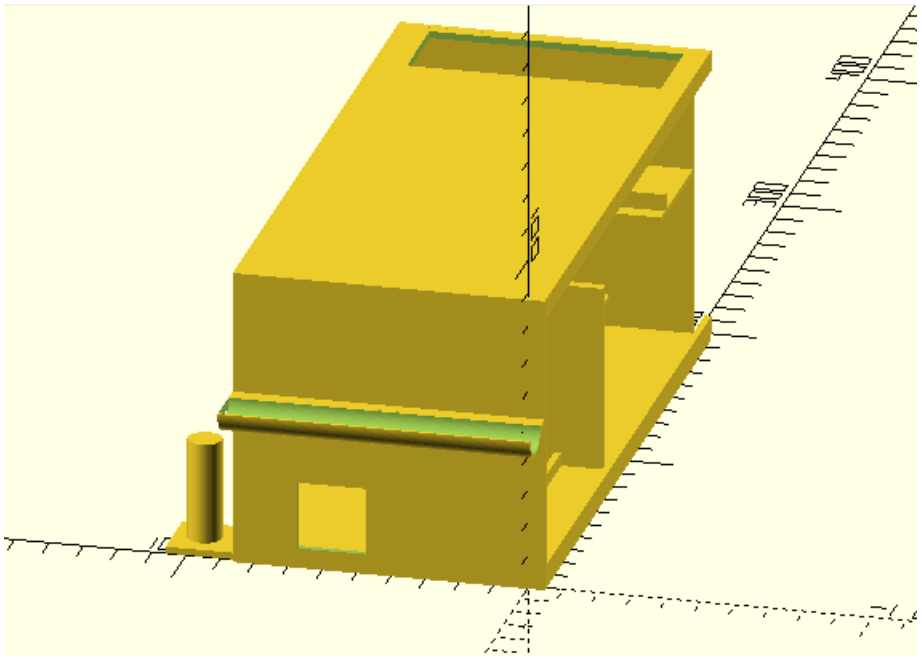
c) The left compartment houses the Arduino Uno, which is held securely in place by a raised bump.



d) **Houses the main perfboard assembly. The upper face directly above this compartment features a rectangular cutout designed specifically for LCD display visibility and visual output monitoring.**

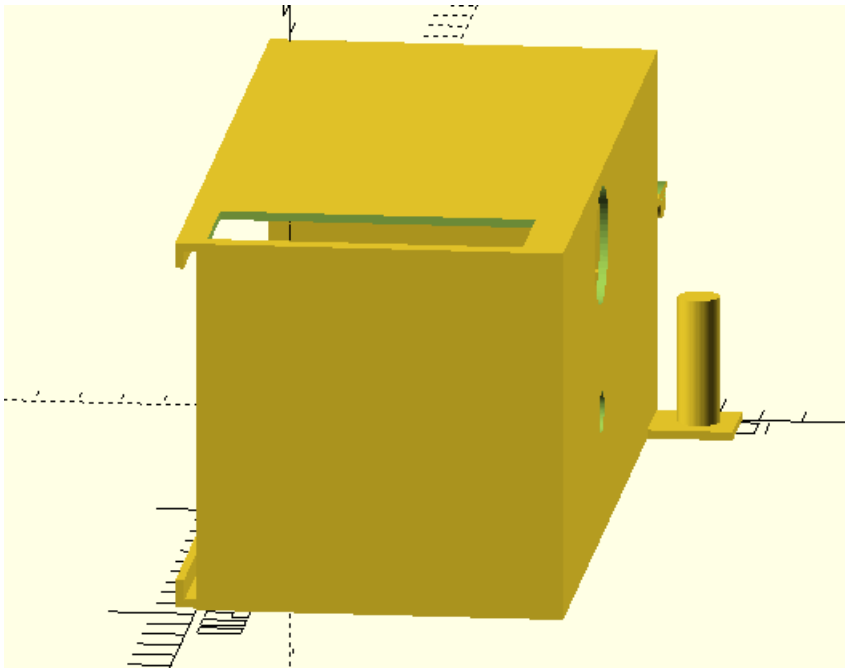


e) **A cylindrical rod was provided at the rear-left side to coil and store the excess PT100 wire. Additionally, an extension featuring a semicylindrical hollow support was provided at the centre of the left face to securely hold the PT100 rod.**

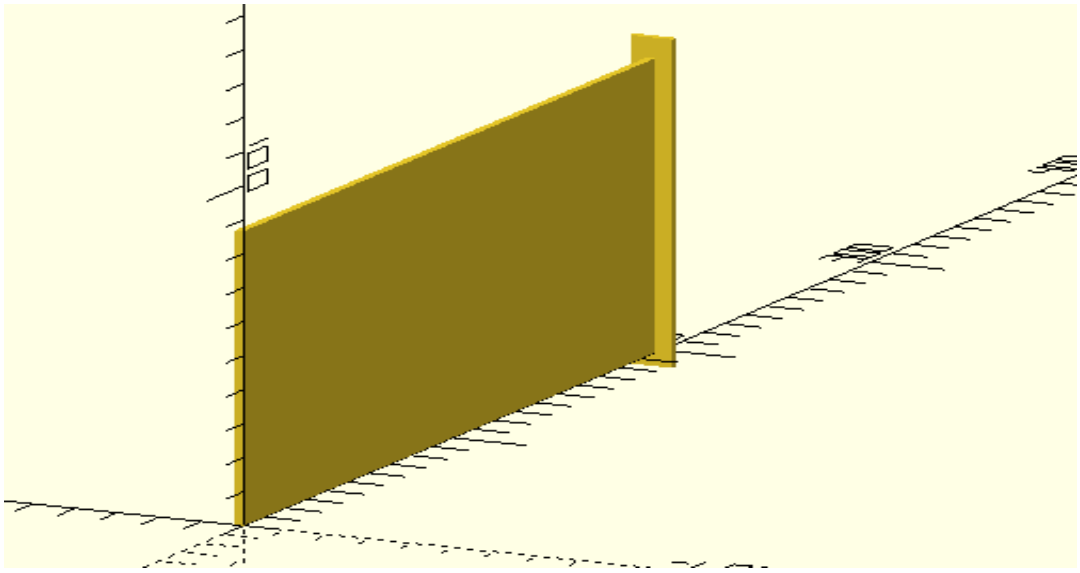


f) Supports were provided on both the left and right sides to securely hold the perf board components in place and ensure proper stability.

g) Extensions were provided along adjacent sides of the front face to create grooves for smoothly sliding the door into place.



i) A small block was provided on the sliding door to serve as a handle, making it easier to open and close the door.



4. References:

a) prof. G. V. V Sharma (pt100 pdf)

b) ChatGPT, Gemini.

Chatgpt – documentation (just for making our sentences with proper vocabulary)